

17 Pythagoras

Materials required: Calculators and rulers

Warm-up

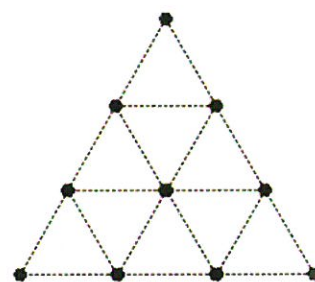
- | | |
|------------------|-------------------|
| 1. 3^2 | 2. 8^2 |
| 3. 2^2 | 4. 11^2 |
| 5. 9^2 | 6. $\sqrt{16}$ |
| 7. $\sqrt{100}$ | 8. $\sqrt{1}$ |
| 9. $\sqrt{2500}$ | 10. $\sqrt{4900}$ |

Discussion

Pythagoras of Samos was an Ionian Greek philosopher, mathematician and founder of the religious movement called Pythagoreanism. He was born on the island of Samos, and it is believed he travelled widely in his youth, visiting Egypt and other places seeking knowledge. He is said to have died in Metapontum.

He is credited with three major discoveries in mathematics, though some believe that he never discovered them himself but that his followers may have made the discoveries. First is Pythagoras' Theorem of the relationship between the side lengths in a right-angled triangle. Second, that musical notes can be translated into mathematical equations and he may be responsible for discovering the properties of string length.

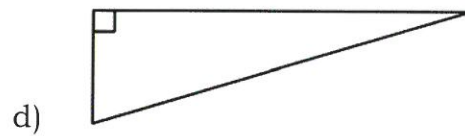
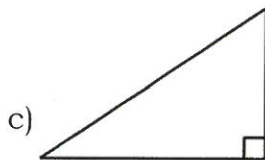
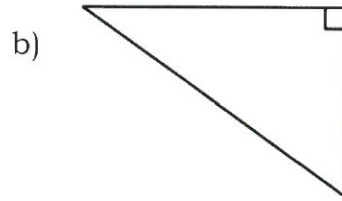
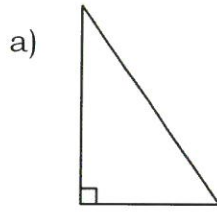
Lastly, Pythagoras was also credited with devising the tetractys, the triangular figure of four rows which add up to the perfect number, ten. As a mystical symbol, it was very important to the worship of the Pythagoreans who would swear oaths by it.



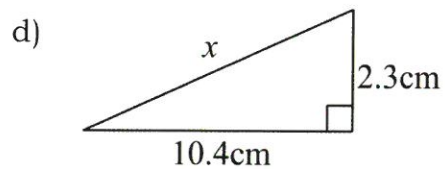
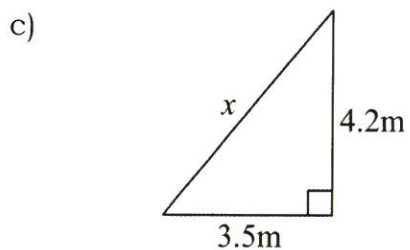
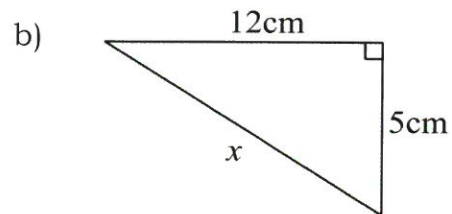
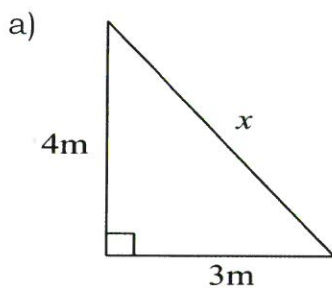
Exploration: See what you can find out about Pythagoras of Samos, the stories of his religion and followers (what they believed and what they did) and anything else interesting about him, them or their actions. Hint: see if you can find out what they thought of beans, or how Pythagoras died.

Exercises

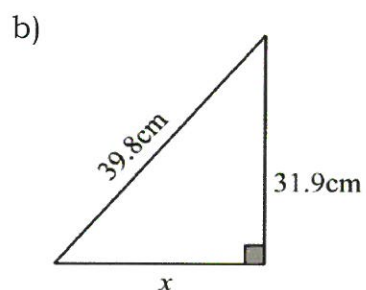
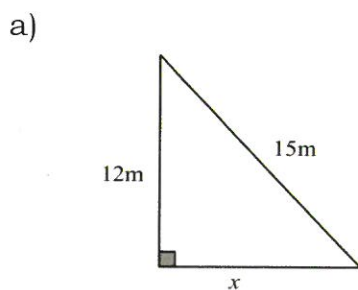
1. Label the hypotenuse in each of the following triangles.



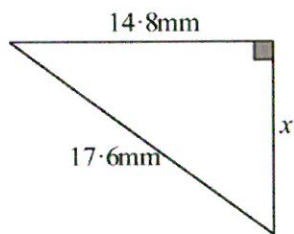
2. Find the length of the hypotenuse to one decimal place, if appropriate.



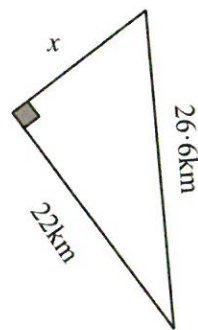
3. Find the length of the shorter side giving answers correct to two decimal places.



c)

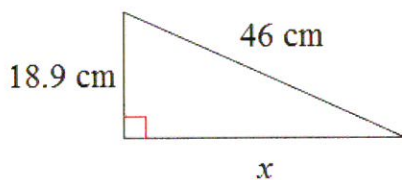


d)

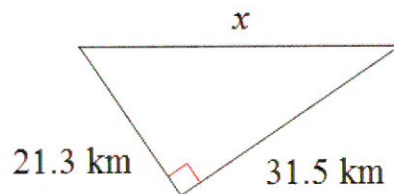


4. Find the length of the unknown side, giving your answers correct to one decimal place.

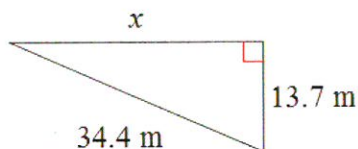
a)



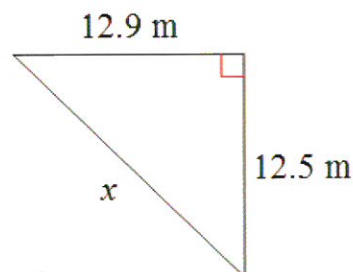
b)



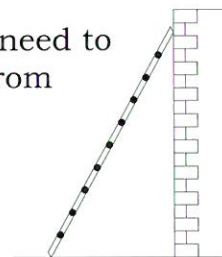
c)



d)



5. A rectangular gate that is 1.8m high and 2.2m wide needs to be repaired using a single brace, a straight support piece fixed across opposite corners of the gate. Determine how long the brace will be for this gate. Hint: draw a sketch to assist.
6. How far from the base of the wall does a 3.2m ladder need to be placed so that the ladder can reach a point 2.9m from the ground up the wall?



Extended Problem Solving

Many trades people use Pythagoras' Theorem to determine if walls, paving, door frames, window frames or brickwork are square, that is at right angles.

One method is to use a version of the 3m, 4m, 5m right angled triangle. Find a suitable corner within your classroom, such as a door frame or window frame. Measure 30cm along one edge and 40cm along the other edge. If the diagonal that joins these two points is 50cm, then the item is square.

Check at least three other items in your classroom.

“Mathematics expresses values that reflect the cosmos, including orderliness, balance, harmony, logic, and abstract beauty.”

— Deepak Chopra

“With me, everything turns into mathematics.”

— René Descartes